

Predicting the Sales Price of Homes in Ames, Iowa

Introduction

To ensure that Century 21 Ames successfully sell all their houses in North Ames (NAmes), Edwards, and Brookside (BrkSide) neighborhoods, we must get an estimate of how the sale price of the house is related to the square footage of the living area of the house and if the sale price and its relationship to square footage depends on which neighborhood the house is located in. Finally, we must build a predictive model for sale prices of homes in all of Ames, Iowa. That way, Century 21 Ames can plan on pricing their homes at a reasonable yet convincing rate for homeowners to consider when scouting their homes.

Notes on Structure

Due to the strict seven-page limit for this assignment, only the most essential tables and figures are included in the main text. Additional supporting tables and figures are provided in the appendix. These are labeled using the prefix “A” (e.g., Table A.1, Figure A.1).

Data Description

The Ames Housing training and testing datasets have been created and collected by Dean De Cock to be used in the data science field. The training set contains 1,460 rows/observations by 81 columns, while the testing set contains 1,459 rows/observations by 80 columns. The difference in the number of columns stems from us making the predicted sale prices of houses in the testing set after constructing the predictive model for sale prices through the training set. Seeing that, the specific variables that we need to know with respect to our analysis are `SalePrice`, `GrLivArea`, `Neighborhood`, `TotalBsmtSF`, `1stFlrSF`, `TotalSF`, `OverallQual`, `YearBuilt`, and `YearRemodAdd`.

1.1: Restatement of Problem

To get an estimate of how the sale price of the house is related to the square footage of the living area of the house, regardless of which neighborhood the house is located in, we have constructed several linear regression models comparing the two variables with 95 percent confidence and prediction intervals, such as a simple linear regression model, a log-linear regression model, a linear-log regression model, and a log-log regression model. After narrowing down which regression model will be the best for our scenario, we use that model to see whether the relationship between the sale price and square footage of houses depends on which neighborhood they are in.

1.2: Comparing Competing Models

To determine whether a log transformation is needed for the model or not, we begin with constructing a simple linear regression model between the sale prices and the square footage of houses in Ames, Iowa. Referring to **Figure A.1**, we discover that a curvature is present in the scatterplot due to numerous outliers, the residual plot displays a clump of data that eventually fans out, and the histogram of residuals displays a relatively normal distribution. Comparing its studentized residual plot to its leverage plot, large studentized residuals and high leverage are present, thus making some

data points influential towards this model, which can be seen from the few spikes in its Cook's D plot. Although there are a few spikes present, there is one extremely large spike at observation 1350 (hits above 1.5), thus placing this observation as extremely influential towards this model. Additionally, it is estimated that 50.18% of the variation in the sale price of a house is explained by its relationship with the square footage of a house. Not only that, but its internal CV PRESS value of 4.63182E12 is an extremely large value, indicating that the predictive accuracy from this model will be limited, and its AIC value of 33392, also being an extremely large value, further suggests that this model is not going to predict efficiently. Therefore, with most of the patterns from these plots being far from ideal along with a relatively low adjusted R-squared value and extremely high internal CV PRESS and AIC values, the simple linear regression model is not an effective model to use to accurately predict the sale prices of houses in Ames, Iowa.

Since the simple linear regression model is far from the ideal model we want to effectively predict the sale prices of houses, we must perform a log transformation to our model. Therefore, we first perform a log-linear transformation on our model. Referring to **Figure A.2**, we discover that a curvature is still present in the scatterplot due to numerous outliers, and the residual plot still displays a clump of data that eventually fans out, but the histogram of residuals now displays a left-skewed distribution. Comparing its studentized residual plot to its leverage plot, most studentized residuals are moderate, and most leverage values are extremely low, indicating that the model is stable for most data points. However, one data point has high leverage and a very large, negative studentized residual (both occurring at a RStudent value greater than -7.5), thus making it influential to this model. This pattern can be seen from the lack of numerous spikes in its Cook's D plot as only two large spikes are present. The larger of the two occurs at observation 1350 (hits close to 1.5), thus placing this observation as extremely influential towards this model. Additionally, it is estimated that only 49.09% of the variation in the sale price of a house is explained by its relationship with the square footage of a house, a percent less than before. Not only that, but its internal CV PRESS value of 119.34867 is a much smaller value than before, indicating that the predictive accuracy from this model is better, and its AIC value of -2,201.37751, an extremely small value, further suggests that this model is going to predict more efficiently than before. Although the outcomes from the log-linear regression model are more promising than what we got from the simple linear regression model, such as more ideal patterns from the plots and low internal CV PRESS and AIC values, the numerous outliers, left-skewed distribution from the histogram of residuals, and the lower adjusted R-squared value are enough evidence to conclude that the log-linear regression model is not the ideal model to effectively predict the sale prices of houses in Ames, Iowa.

Since the log-linear regression model is still far from the ideal model we want to effectively predict the sale prices of houses, we then perform a linear-log transformation on our model. Referring to **Figure A.3**, we discovered that a curvature is present in both the scatterplot and residual plot due to numerous outliers, and the histogram of residuals displays a slightly right-skewed distribution. Comparing its studentized residual plot to its leverage plot, moderate studentized residuals and leverage are mostly present; there are no points that are both high-leverage and high-residual, thus, there are no influential data points nor dangerous outliers present in this model. This pattern can be seen in its Cook's D plot as all the larger spikes shown are less than 1. Additionally, it is estimated that only 48.28% of the variation in the sale price of a house is explained by its relationship with the square footage of a house, a percent less than before. Not only that, but its internal CV PRESS value of 4.781641E12 is an extremely large value (similar to the one in the simple linear regression model), indicating that the predictive accuracy from this model will be limited, and its AIC value of 33447, also being an extremely large value (and similar to the one in the simple linear regression model), further suggests that this model is not going to predict efficiently. Although the outcomes from the linear-log regression model are still more promising than what we got from the simple linear regression model, such

as the patterns found from the studentized residuals, leverage, and Cook's D plots, the curvature pattern, right-skewed distribution from the histogram of residuals, the lower adjusted R-squared value, and larger internal CV PRESS value and AIC value are more than enough evidence to conclude that the linear-log regression model is not the ideal model to effectively predict the sale prices of houses in Ames, Iowa.

Since the linear-log regression model is also still far from the ideal model we want to effectively predict the sale prices of houses, we then perform a log-log transformation on our model. Referring to **Figure A.4**, we discovered that a linear relationship is present in the scatterplot, a random cloud is present in the residual plot, and the histogram of residuals displays a relatively normal distribution. Comparing its studentized residual plot to its leverage plot, the studentized residuals are moderate, and leverage is low to moderate; there are no points that are both high-leverage and high-residual, thus, there are no influential data points nor dangerous outliers present in this model. This pattern can be seen in its Cook's D plot as the few larger spikes shown are all less than 1. Additionally, it is estimated that 53.3% of the variation in the sale price of a house is explained by its relationship with the square footage of a house, much greater than any of the models we constructed before. Not only that, but its internal CV PRESS value of 109.02025 is the smallest value than any of the other internal CV PRESS values, indicating that the predictive accuracy from this model is going to be the strongest of all, and its AIC value of -2,327.10645, also the smallest value than any of the models we constructed before, further suggests that this model is going to be the most efficient at predicting. With all this, the ideal patterns seen from the plots, the highest adjusted R-square value, and the extremely small internal CV PRESS value and AIC value make the log-log regression model the best model to effectively predict the sale prices of houses in Ames, Iowa.

1.3: Parameters of the Best Model

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate	Variance Inflation
Intercept	1	5.66812	0.15589	36.36	<.0001	0	0
lGrLivArea	1	0.87454	0.02143	40.82	<.0001	0.73025	1.00000

$$\log(\widehat{SalePrice}) = 5.66812 + 0.87454 * \log(GrLivArea)$$

Figure 1: Table of parameter estimates, 95% Confidence Intervals, and best estimate of the correlation between $\log(GrLivArea)$ and $\log(SalePrice)$.

The analysis supports that log-log transformation is the best model. By the equation in **Figure 1**, a doubling of the square footage of the living space of houses is associated with a multiplicative change of $2^{0.87454} = 1.833$ in the median of the distribution of sale prices of houses. In other words, a doubling of the square footage of the living space of houses increases the estimated median of sale prices of houses by 83.3%. The 95% confidence interval for the multiplicative increase in the median of sale prices of houses after a doubling of the square footage of the living space of houses is $(2^{0.8325}, 2^{0.9166}) = (1.78, 1.89) = (78\%, 89\%)$ increase.

1.4: Sale Price vs. Sq Footage Adjusted by Neighborhood

To see whether the sale price and its relationship to square footage depends on which neighborhood a house is located at, we focus on houses from three neighborhoods: North Ames (NAMES), Edwards, and Brookside (BrkSide). Based on the scatterplot shown in **Figure 2**, we notice that most houses located in the North Ames neighborhood (light green data points) are generally sold at higher prices for the same square footage of the living space of houses as the other two neighborhoods. However, there are a few houses located in the North Ames neighborhood that have larger square footage in relation to its higher prices. Additionally, we notice that houses located in the

Edwards neighborhood (navy blue data points) are generally sold at slightly lower prices for the same square footage of the living space of houses as the other two neighborhoods. There is moderate pricing relative to the square footages of houses in the Edwards neighborhood. Finally, we notice that houses located in the Brookside neighborhood (dark green data points) are sold at the most affordable prices for the same square footage of the living space of houses as the other two neighborhoods. However, there are a few houses located in the Brookside neighborhood that have smaller square footage in relation to its more affordable prices. Therefore, we can conclude that the sale price and its relationship to square footage does depend on which neighborhood a house is located at.

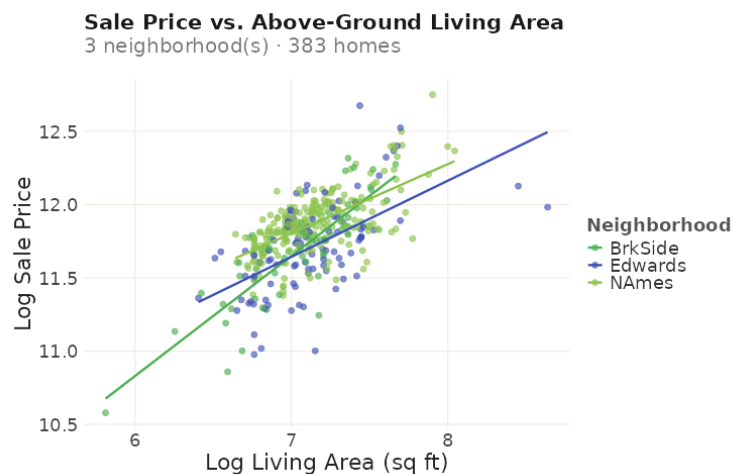


Figure 2: Scatterplot of Sale Price vs. Sq Footage of Houses in North Ames, Edwards, and Brookside Neighborhoods. [Explore more on the RShiny App.](#)

1.5: Intermittent Conclusion

As shown in **Figure 1**, there is very strong evidence at the $\alpha = 0.05$ level of significance to suggest that the log of square footage of the living space of a house is linearly related to the log of sale price of a house in all neighborhoods of Ames, Iowa (p-value of < 0.0001). It is estimated that 53.3% of the variation in the sale price of a house is explained by its relationship with the square footage of a house based on the log-log regression model. Additionally, comparing the three specific neighborhoods in Ames, Iowa, based on which neighborhood a house is located at determines how much square footage it should have along with the price it should be marketed at. If one wants large square footage of the living space of a house and is willing to pay the higher cost, one should live in the North Ames neighborhood. However, if one wants moderate square footage of the living space of a house with a more reasonable price, one should live in the Edwards neighborhood. Finally, if one cannot afford to live in the North Ames nor the Edwards neighborhoods, one can live in a house with smaller square footage in the Brookside neighborhood.

2.1: Predictive Models for Sale Price across Ames, Iowa

The next phase involved developing a series of predictive models for housing prices across the city of Ames, Iowa. This process had three primary objectives. First, a simple model was constructed using a single explanatory model. This model prioritizes transparency and ease of explanation. Second, a complex model was constructed to optimize predictive accuracy within the constraints of ordinary least squares (OLS). This approach involved feature engineering with no restriction on the number of parameters. This model scores well on the test set, but the complexity limits its interpretability and raises questions about overfitting. Finally, the most important features from the full model were selected for a reduced model that balances interpretability with test set performance. In the course of model development, outliers were managed, assumptions were checked, and features were engineered to

better capture underlying relationships within the data. The three models are compared based on predictive accuracy, interpretability, and susceptibility to overfitting.

2.2: The Simple Model

Top 10 Correlations with SalePrice					
Pearson's R — Numeric Variables					
Rank	Variable	Pearson's R	Rank	Variable	Pearson's R
1	OverallQual	0.7910	6	1stFlrSF	0.6059
2	GrLivArea	0.7086	7	FullBath	0.5607
3	GarageCars	0.6404	8	TotRmsAbvGrd	0.5337
4	GarageArea	0.6234	9	YearBuilt	0.5229
5	TotalBsmtSF	0.6136	10	YearRemodAdd	0.5071

Table 1: Top ranked correlations between `SalePrice` and other numeric variables.

The first step was to identify the most suitable explanatory variable for the simple model. During the initial exploratory analysis, Pearson's correlation coefficient (r) was computed between each numeric feature and `SalePrice`. As shown in **Table 1**, the strongest correlation was observed with `OverallQual`. However, this variable is less desirable from an interpretability standpoint because its meaning is not transparent. According to the dataset documentation, `OverallQual` represents a composite assessment of overall material quality and finish, but the exact scoring criteria are not clearly defined and are likely subjective.

The second strongest correlation was found with `GrLivArea`, which offers greater interpretability. It is based on a direct, objective measure of above-ground living space. Additionally, other area-based features, `TotalBsmtSF` and `1stFlrSF`, also rank among the top ten. Taken together with common sense, these results suggest that total living space is a key driver of housing prices. Accordingly, model development proceeded with a focus on interpretable, area-based measures.

$$TotalSF = TotalBsmtSF + 1stFlrSF + 2ndFlrSF$$

Equation 1: Engineered feature, `TotalSF` (Total Square Feet), which combines the square footage of the basement with the first and second floors.

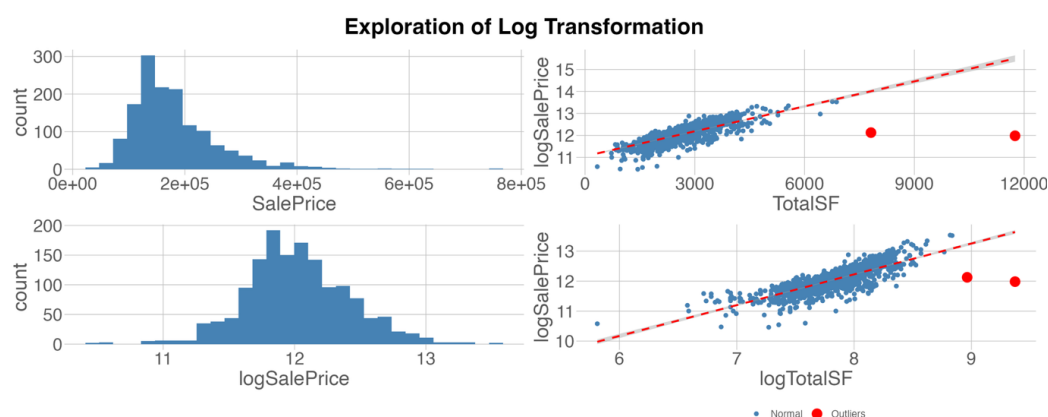


Figure 3: Impact of log transformation on the correlation between `SalePrice` and `TotalSF`. `SalePrice` is distinctly right skewed, and with log transformation, the distribution normalizes. Both natural and log-scaled `TotalSF` are satisfactorily linear in relation to $\log(SalePrice)$. However, log transformation moderates the leverage of outliers, and the R^2 value is superior. These results are consistent with the analysis of `GrLivArea`.

$$\log(\widehat{\text{SalePrice}}) = 3.787 + 1.056 * \log(\text{TotalSF})$$

Equation 2: Best fit model to predict *SalePrice* given *TotalSF* (OLS with outliers excluded, $R^2=0.6688$).

Through trial and error, *TotalSF* (**Equation 2**) was found to be the best area-based predictor of *SalePrice* (ANOVA, $p < .0001$). A doubling of *TotalSF* is associated with a 103.9% increase in predicted *SalePrice* (95% CI: {98.6%, 109.5%}). By the R^2 , the model is estimated to explain 65.14% of variance of the $\log(\text{Sale Price})$. For a single engineered variable out of 79 features, this is a respectable starting point. **Figure 3** highlights a pair of outliers with exceptionally high square footage and a low sale price. The outliers are excluded by limiting the domain of generalization to a maximum *TotalSF* of 7,500 sq. feet. After recalibrating the model, a doubling of *TotalSF* is associated with a 107.9% in predicted *SalePrice* (95%CI: {102.5%, 113.5%}), and the R^2 climbs to 0.6688. The intercept is estimated to be 3.787 (95% CI: {3.489, 4.086}), but this is extrapolation beyond the observed range and has no practical significance. The two outliers were investigated features that might distinguish them from the general population. Evidence was found in the attribute, *SaleCondition*. Over 90% of the dataset was marked at “Typical”, but these two listings were recorded as “Partial”. The meaning of “Partial” is not fully understood, but it supports that these two listings are separate from the general population. In all models that follow, the outliers are excluded from the train set.

2.3: The Full and Reduced Models

Building on the simple model, a more comprehensive modeling approach was developed to improve predictive performance while maintaining interpretability. Exploration and feature engineering were used to identify and construct the most informative predictors. The final set of candidate features, along with their categorizations, are summarized in **Figure A.5**.

Several key patterns emerged from this process. First, more granular measures of property size improved predictive power beyond *TotalSF* alone. In particular, features such as lot area and porch size captured additional variation in property value that is not reflected in interior square footage. These variables help account for outdoor space and usability, which are important considerations in housing markets. Second, build quality remained a strong determinant of *SalePrice*. *OverallQual* was found to be the most important explanatory variable. More granular features, such roof material and the presence of masonry features provided more transparent indicators of construction quality and also contributed meaningfully to model performance. Third, age-related variables were consistently important. Both *YearBuilt* and *YearRemodAdd* capture different dimensions of property age. These variables help distinguish between older homes that have been updated and those that have not, which has a clear impact on market value. Finally, location-based factors also played a significant role. Zoning classifications, in particular, were found to influence *SalePrice*, likely reflecting differences in land use, neighborhood characteristics, and development restrictions.

The reduced model was developed by pruning features from the full model and achieves comparable performance with less than half the number of parameters. The largest reduction comes from clustering neighborhoods. The dataset includes 25 neighborhoods. In the full model, each is represented once as an intercept adjustment and twice more through interactions with *TotalSF* and *OverallQual*. Consequently, nearly half of the approximately 150 parameters are tied to neighborhood effects. While allocating three parameters per neighborhood is reasonable in most cases, the bottom third of neighborhoods contain fewer than 30 observations and are better suited for grouping. To address this, k-means clustering was applied using mean *SalePrice*, reducing the 25 neighborhoods to 10 groups and eliminating 45 parameters. **Figure 4** illustrates this clustering by comparing neighborhoods based on mean *SalePrice*.

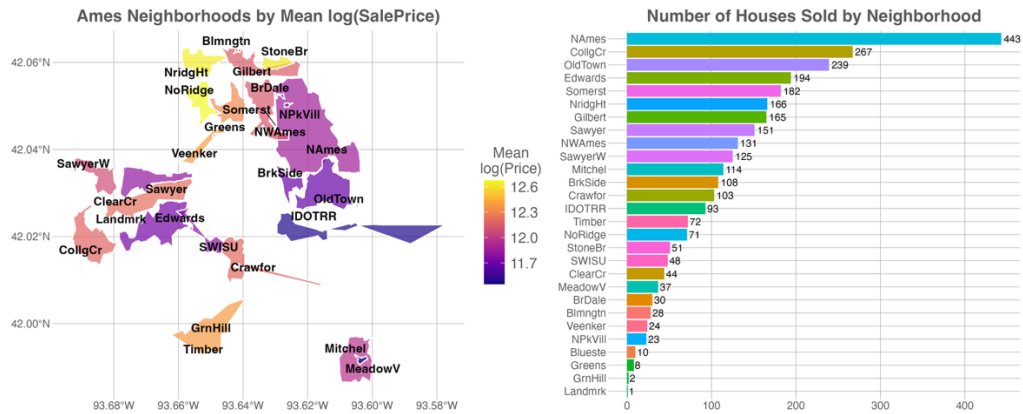


Figure 4: Heatmap of mean (SalePrice) by Neighborhood (left) and barchart of houses sold by Neighborhood.

2.4: Checking the Assumptions

Figure A.6 presents diagnostic plots for both the full and reduced models. The Q–Q plots and histograms of the residuals indicate that the residuals are approximately normally distributed. The plot of studentized residuals shows a random scatter, suggesting that this normality holds across the observed range. Cook’s D values peak at 0.06, indicating that no individual observation exerts undue influence on the model.

2.4: Results and Conclusion

Predictive Model Comparison						
Predictive Models	Parameters	Adjusted R ²	CV PRESS	AIC	RMSE (\$)	Kaggle Score
Simple	2	0.6686	77.40	−142.5	45,259	0.22896
Reduced	56	0.9136	21.43	−2,050.0	21,812	0.13567
Full	120	0.9314	17.89	−2,328.1	18,660	0.13339

Table 2: Performance metrics of the Simple, Reduced, and Full models.

Table 2 summarizes the performance of the Simple, Full, and Reduced models. As expected, the Simple model provides strong interpretability but comparatively weaker predictive performance. In contrast, both the Full and Reduced models perform substantially better on the test set and rank competitively, falling within the top 40% of submissions. Notably, the Reduced model achieves performance comparable to the Full model despite using less than half the number of parameters, indicating a more efficient balance between complexity and predictive power. An ANOVA comparison of the Full and Reduced models provides statistical evidence that at least one pairing of grouped neighborhoods differs significantly ($p < 0.0001$). However, this difference is likely not practically significant.

An important caveat Full model was developed through repeated evaluation on the test set. This is problematic, and the full model is likely overfit. Therefore, although the Reduced model underperforms on the test set, there is no evidence to suggest that it would underperform on truly unseen data. Its reduced complexity may, in fact, improve generalization. Taken together, these results suggest that the current models are approaching the practical limits of predictive performance under the assumptions of ordinary least squares (OLS). Further improvements are unlikely to come from additional feature engineering alone. Instead, more advanced methods, such as regularized regression techniques (e.g., Ridge and Lasso) or ensemble-based approaches like XGBoost, may provide meaningful gains in predictive accuracy by capturing more complex relationships within the data.

Appendix

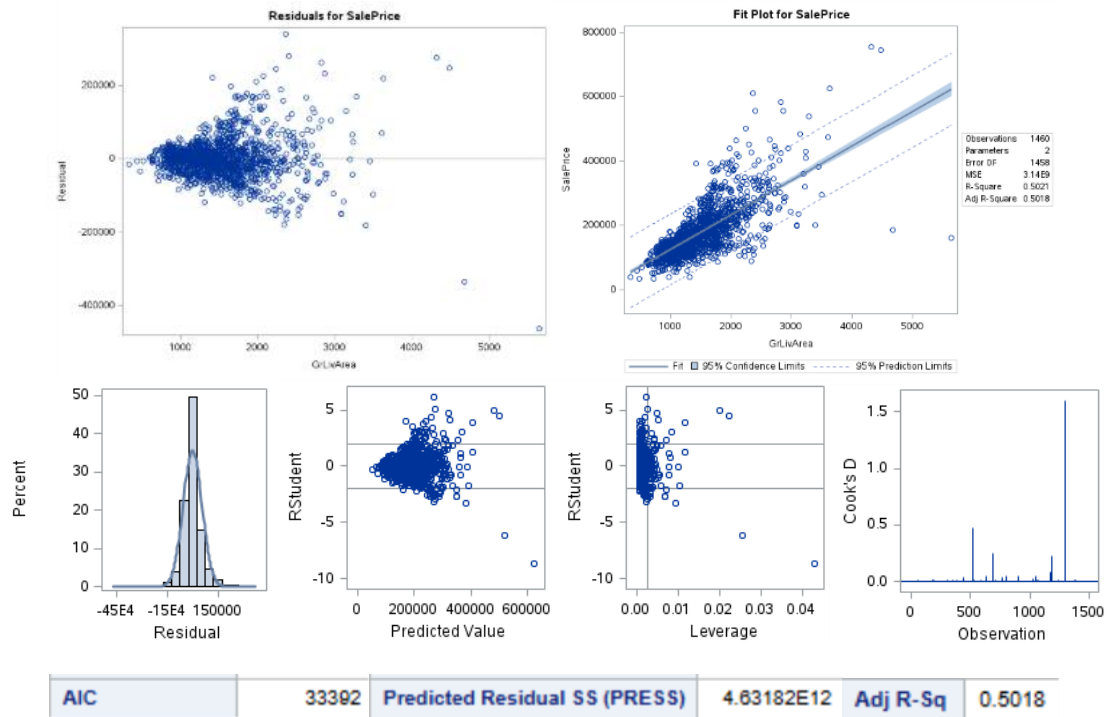


Figure A.1: Outcomes of natural scaled linear regression.

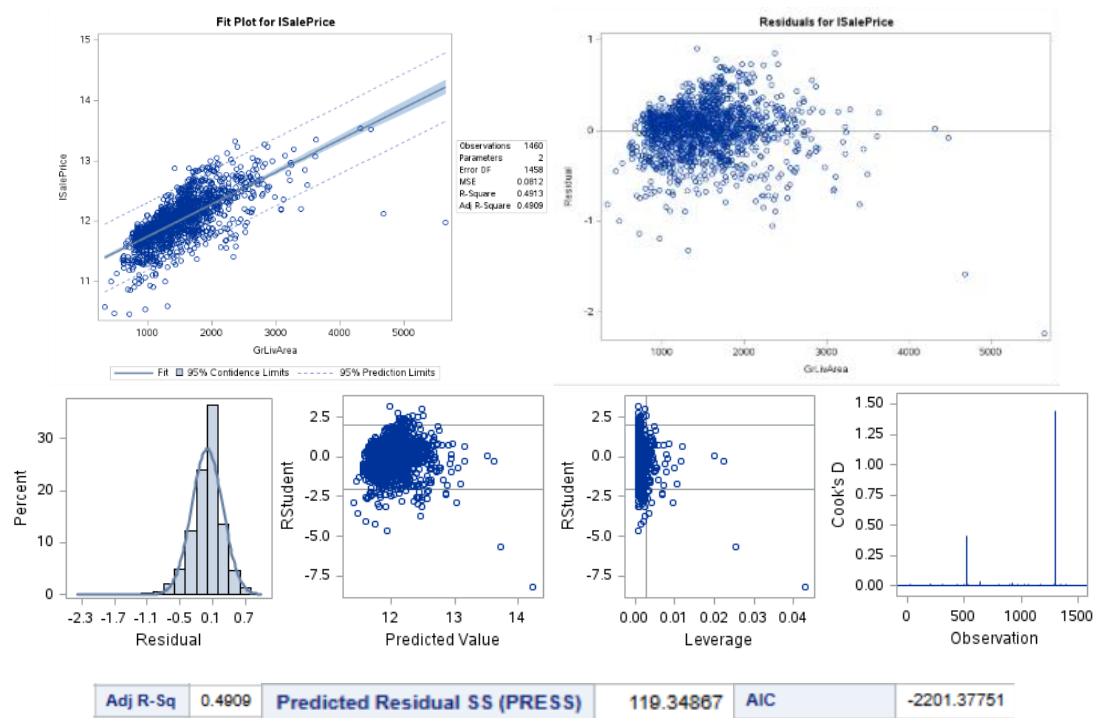


Figure A.2: Outcomes of the log-linear regression model.

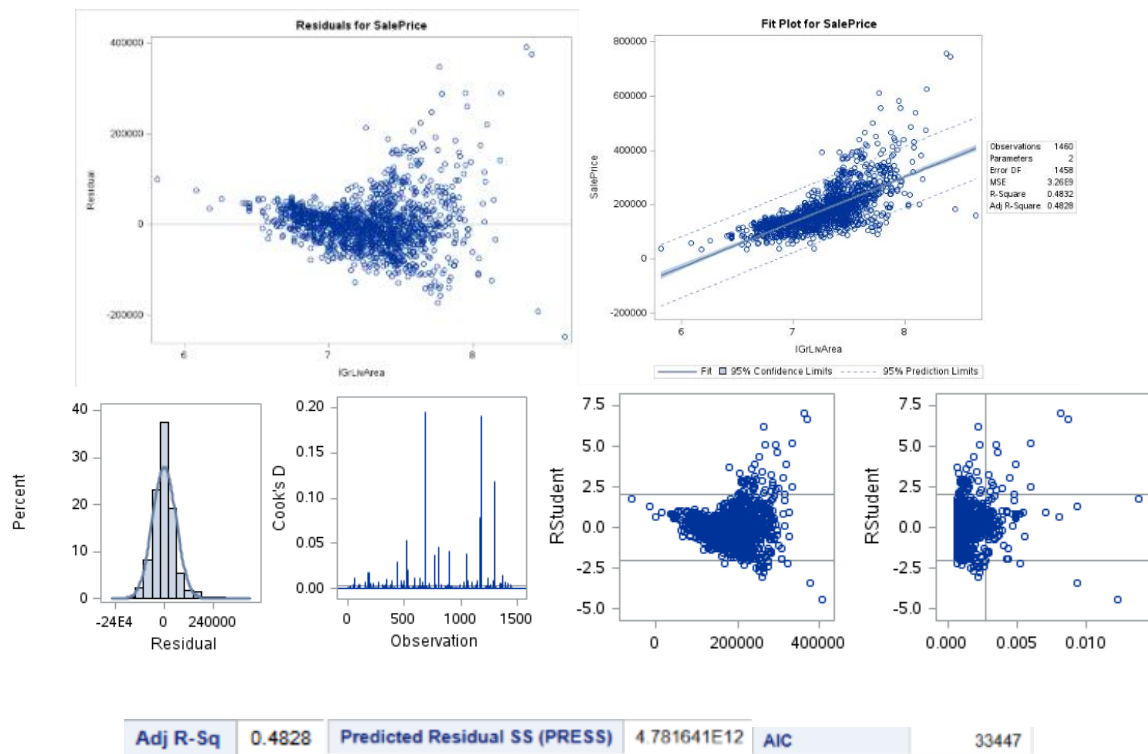


Figure A.3: Outcomes of the linear-log regression model.

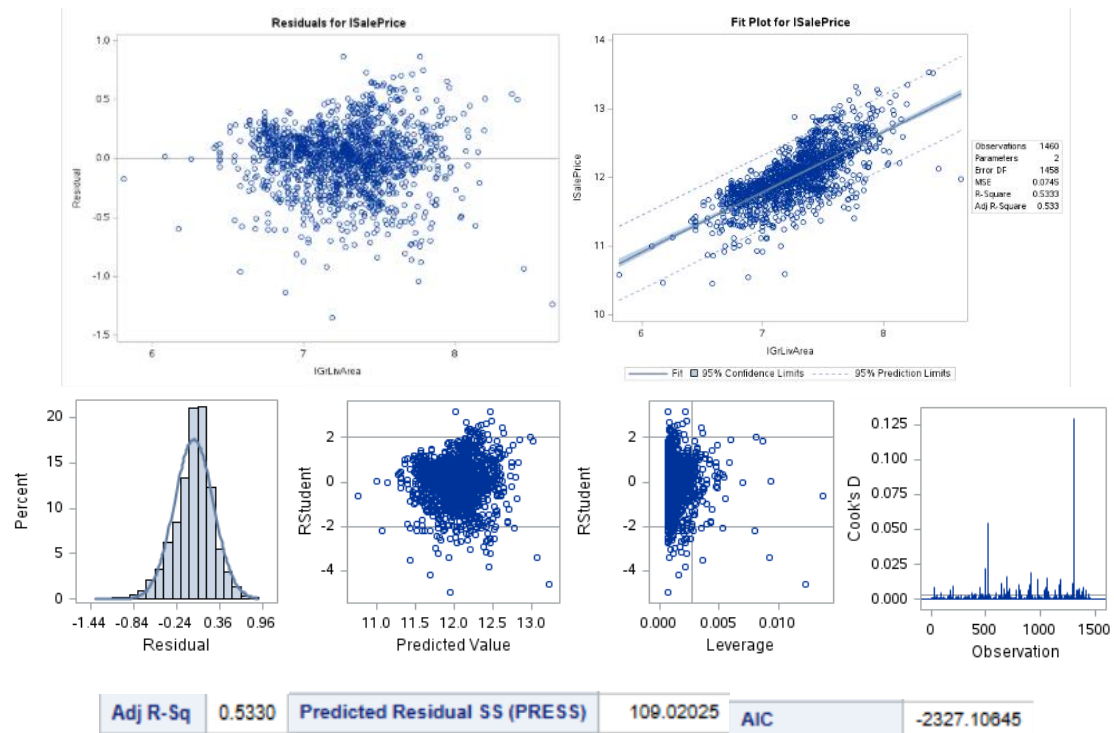


Figure A.4: Outcomes of the log-log regression model.



Figure A.5: Data definitions, flowchart, and code for the simple, full, and reduced models.

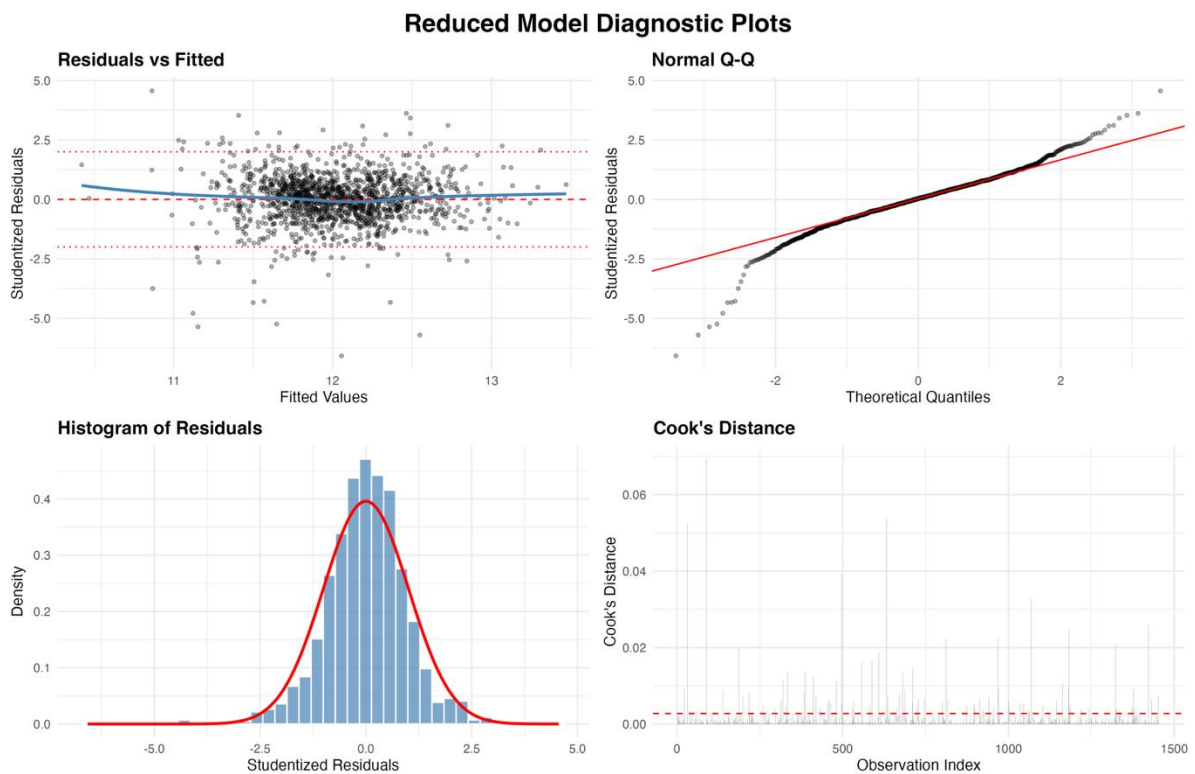
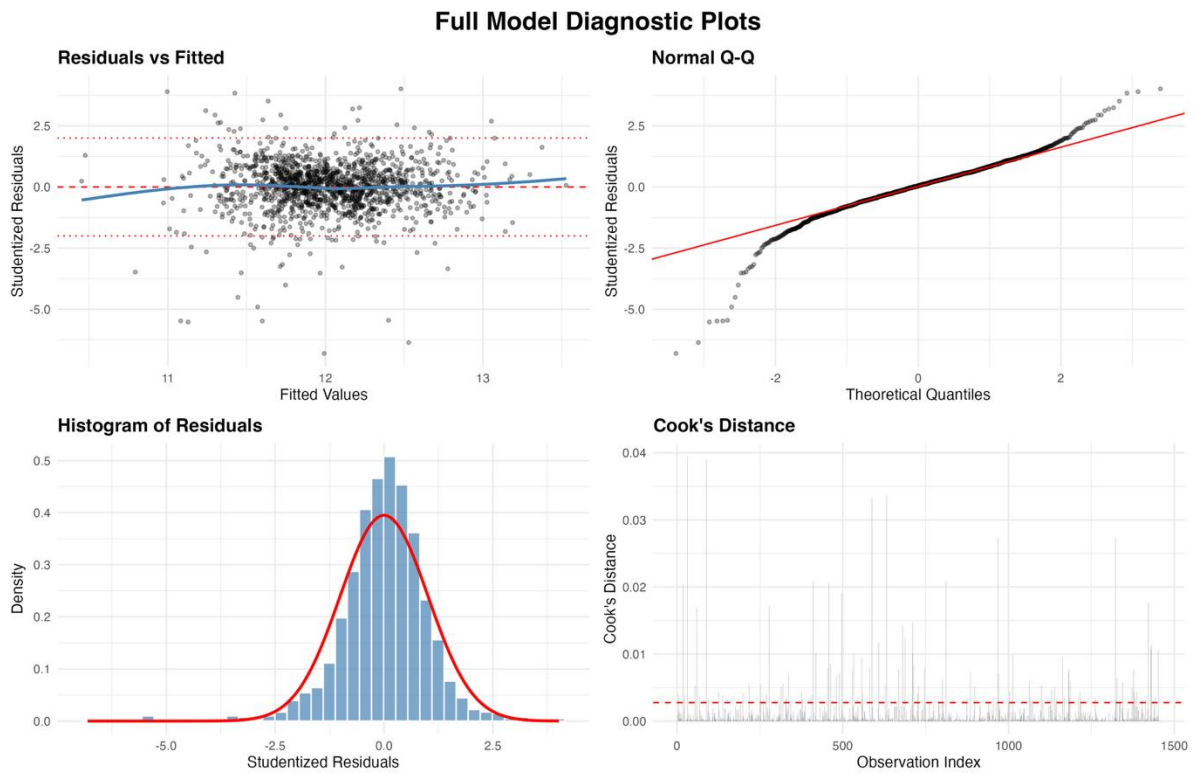


Figure A.6: Diagnostic plots for the full and reduced models.

SAS

```
* import training set;
proc import datafile = '/home/u64273612/sasuser.v94/train.csv'
    out = work.all_data
    dbms = csv
    replace;
    getnames = yes;
run;

* subset just the GrLivArea and SalePrice columns;
data work.subset_data;
    set work.all_data(keep = GrLivArea SalePrice Neighborhood);
run;

proc print data = work.subset_data;
run;

* original linear regression (no transformations have been done);
proc reg data = work.subset_data;
    model SalePrice = GrLivArea / press stb vif influence;
    output out = regout p = pred r = residual press = pressval;
run;
* curvature is present due to ton of outliers in scatterplot;
* residual plot shows data being clumped into one area and then eventually
fanning out;
* relatively normal distribution is present;
* high leverage and large studentized residuals (making points influential);
* few spikes present in Cook's D plot (potentially influential);
* one very large spike in Cook's D plot (> 1) at around 1350 (highly influential);
* adjusted r-squared value = 0.5018 (explains half of the variation);
* PRESS value: 4.63182E12 (a really big value, so predictive accuracy is limited,
therefore making this model struggle to predict new prices for houses well);

* get AIC value for original linear regression (no transformations have been
done);
proc glmselect data = work.subset_data;
    model SalePrice = GrLivArea / selection = none stats = all;
run;
* AIC value: 33392 (high value, thus suggesting that the model isn't the most
efficient one to predict sale prices);

* To take the log of SalePrice (log-linear transformation);
data work.subset_data;
set work.subset_data;
lSalePrice = log(SalePrice); * take the log of Y;
run;

* To check regression assumptions of transformed data and to get the
parameter estimates;
proc reg data = work.subset_data;
model lSalePrice = GrLivArea / press stb vif influence;
    output out = regout p = pred r = residual press = pressval;
run;
* curvature still present due to ton of outliers in scatterplot;
* residual plot shows data being clumped into one area but eventually fans out;
* left-skewed distribution is present in histogram of residuals;
* most leverage values are extremely low and most studentized residuals
are moderate;
* but, one point has high leverage and very large negative studentized residual,
making it an influential outlier;
* two large spikes in Cook's D plot --> larger of the two at around 1350 makes the
data point highly influential;
* adjusted r-squared value = 0.4909 (worse than original LR);
* PRESS value = 119.34867 (much better than before, model struggles less with
predicting);

* get AIC value for original linear regression (log-linear transformation);
```

```

proc glmselect data = work.subset_data;
    model lSalePrice = GrLivArea / selection = none stats = all;
run;
* AIC value: -2201.37751 (very small value, thus suggesting that the model is
better than before);

* To take the log of GrLivArea (linear-log transformation);
data work.subset_data;
set work.subset_data;
lGrLivArea = log(GrLivArea); * take the log of X;
run;

* To check regression assumptions of transformed data and to get the
parameter estimates;
proc reg data = work.subset_data;
model SalePrice = lGrLivArea / press stb vif influence;
    output out = regout p = pred r = residual press = pressval;
run;
* curvature is present in both the scatterplot and residual plot due
to ton of outliers;
* slightly right-skewed distribution for histogram of residuals;
* moderate leverage and studentized residuals mostly
* no point is both high leverage and high residual, so no influential points nor
dangerous outliers, so model is stable;
* few spikes in Cook's D plot, but none of them are greater than 1 (no influential
points);
* adjusted r-squared value = 0.4828 (not good either);
* PRESS value = 4.781641E12 (a really large value, making prediction accuracy
limited, making model struggle to predict);

* get AIC value for original linear regression (linear-log transformation);
proc glmselect data = work.subset_data;
    model SalePrice = lGrLivArea / selection = none stats = all;
run;
* AIC value: 33447 (high value, thus suggesting the model to not be the most
efficient);

* To check regression assumptions of transformed data and to get the parameter
estimates (log-log transformation);
proc reg data = work.subset_data;
model lSalePrice = lGrLivArea / press stb vif influence;
    output out = regout p = pred r = residual press = pressval;
run;
* linear relationship is present;
* most random cloud I have seen in a residual plot;
* relatively normal distribution is present;
* low to moderate leverage and moderate studentized residuals;
* no point is both high leverage and high residual (no influential points nor
dangerous outliers, so model is stable);
* few spikes in Cook's D plot (largest spike is at 0.125, which is still less than
1, so no influential points);
* adjusted r-squared value = 0.5330 which is the best out of all models so far;
* PRESS value = 109.02025 (smallest value of all, model struggles less on
predicting);

* get AIC value for original linear regression (log-log transformation);
proc glmselect data = work.subset_data;
    model lSalePrice = lGrLivArea / selection = none stats = all;
run;
* AIC value: -2327.10645 (very small value, even smaller than the one for log-
linear transformation, best model to use)
* will use log-log transformation model to make my analysis;

* filtering work.subset_data to specific neighborhoods;
proc sgplot data = work.subset_data;
    where Neighborhood in ("NAmes", "Edwards", "BrkSide");
    scatter x = lGrLivArea y = lSalePrice / group = Neighborhood;
run;

```

- * NAmes (red) - sell at higher prices for the same square footage of houses;
- * Edwards (green) - slightly cheaper than NAmes, moderate pricing relative to square footage of houses;
- * BrkSide (blue) - cheapest out of the 3 neighborhoods for the same square footage of houses (lower-valued neighborhood);

R Code

###MAIN

```
library(tidyverse)
library(gt)
library(splines)
library(ggplot2)
library(patchwork)

project <- "/Users/jwkle/Documents/Stats/HW/project/"
path_ <- function(file) paste0(project, file)

# -----
# LOAD
# -----
Train_raw <- read_csv(path_("train.csv"), show_col_types = FALSE)
Test_raw <- read_csv(path_("test.csv"), show_col_types = FALSE)

# -----
# TRAIN-DERIVED IMPUTATION VALUES
# -----
train_mszoning_mode <- names(sort(table(Train_raw$MSZoning), decreasing =
TRUE))[1]
train_salecond_mode <- names(sort(table(Train_raw$SaleCondition), decreasing =
TRUE))[1]
train_kitchenqual_mode <- names(sort(table(Train_raw$KitchenQual), decreasing =
TRUE))[1]
train_bsmtqual_mode <- names(sort(table(Train_raw$BsmtQual), decreasing =
TRUE))[1]
train_bsmtexp_mode <- names(sort(table(Train_raw$BsmtExposure), decreasing =
TRUE))[1]
train_landslope_mode <- names(sort(table(Train_raw$LandSlope), decreasing =
TRUE))[1]
train_condition2_mode <- names(sort(table(Train_raw$Condition2), decreasing =
TRUE))[1]
train_mssubclass_mode <- names(sort(table(Train_raw$MSSubClass), decreasing =
TRUE))[1]
train_extqual_mode <- names(sort(table(Train_raw$ExterQual), decreasing =
TRUE))[1]
train_functional_mode <- names(sort(table(Train_raw$Functional), decreasing =
TRUE))[1]
train_garagefinish_mode <- names(sort(table(Train_raw$GarageFinish), decreasing =
TRUE))[1]

train_lotarea_med <- median(Train_raw$LotArea, na.rm = TRUE)
train_masvnrarea_med <- median(Train_raw$MasVnrArea, na.rm = TRUE)
train_bsmtfinsf1_med <- median(Train_raw$BsmtFinSF1, na.rm = TRUE)
train_bsmtfinsf2_med <- median(Train_raw$BsmtFinSF2, na.rm = TRUE)
train_bsmtunfsf_med <- median(Train_raw$BsmtUnfSF, na.rm = TRUE)
train_firstflrsf_med <- median(Train_raw$`1stFlrSF`, na.rm = TRUE)
train_secondflrsf_med <- median(Train_raw$`2ndFlrSF`, na.rm = TRUE)
train_bedroomabvgr_med <- median(Train_raw$BedroomAbvGr, na.rm = TRUE)
train_garagearea_med <- median(Train_raw$GarageArea, na.rm = TRUE)
train_poolarea_med <- median(Train_raw$PoolArea, na.rm = TRUE)
train_grlivarea_med <- median(Train_raw$GrLivArea, na.rm = TRUE)
train_kitchenabvgr_med <- median(Train_raw$KitchenAbvGr, na.rm = TRUE)
train_lotfrontage_med <- median(Train_raw$LotFrontage, na.rm = TRUE)
train_wooddecksf_med <- median(Train_raw$WoodDeckSF, na.rm = TRUE)
train_openporchsf_med <- median(Train_raw$OpenPorchSF, na.rm = TRUE)
train_garageyrblt_med <- median(Train_raw$GarageYrBlt, na.rm = TRUE)

# -----
# NEIGHBORHOOD LOOKUP
# -----
neigh_base <- Train_raw %>%
  mutate(log_price = log(SalePrice)) %>%
```

```

group_by(Neighborhood) %>%
  summarise(mean_log_price = mean(log_price, na.rm = TRUE), .groups = "drop")

set.seed(123)
km <- kmeans(neigh_base[, "mean_log_price", drop = FALSE], centers = 10, nstart =
25)
neigh_base$cluster <- km$cluster

cluster_order <- neigh_base %>%
  group_by(cluster) %>%
  summarise(cluster_mean = mean(mean_log_price), .groups = "drop") %>%
  arrange(cluster_mean)

label_map <- setNames(paste0("Group_", 1:10), cluster_order$cluster)
neigh_base$NeighborhoodGroup <- factor(label_map[as.character(neigh_base$cluster)],
levels = paste0("Group_", 1:10))
neigh_lookup <- neigh_base %>% dplyr::select(Neighborhood, NeighborhoodGroup)

# -----
# ENGINEER
# -----
engineer <- function(df) {
  df %>%
    mutate(
      FunctionalRed = as.factor(if_else(Functional == "Typ", "Typ", "Deduc")),
      TotalBsmtSF = replace_na(TotalBsmtSF, 0),
      BsmtFullBath = replace_na(BsmtFullBath, 0),
      BsmtHalfBath = replace_na(BsmtHalfBath, 0),
      GarageCars = replace_na(GarageCars, 0),
      LotFrontage = replace_na(LotFrontage, train_lotfrontage_med),
      LotArea = replace_na(LotArea, train_lotarea_med),
      BsmtUnfSF = replace_na(BsmtUnfSF, train_bsmtunfsf_med),
      GrLivArea = replace_na(GrLivArea, train_grlivarea_med),
      KitchenAbvGr = replace_na(KitchenAbvGr, train_kitchenabvgr_med),
      ScreenPorch = replace_na(ScreenPorch, 0),
      WoodDeckSF = replace_na(WoodDeckSF, 0),
      OpenPorchSF = replace_na(OpenPorchSF, 0),
      MasVnrArea = replace_na(MasVnrArea, train_masvnrarea_med),
      BsmtFinSF1 = replace_na(BsmtFinSF1, train_bsmtfinsf1_med),
      BsmtFinSF2 = replace_na(BsmtFinSF2, train_bsmtfinsf2_med),
      `1stFlrSF` = replace_na(`1stFlrSF`, train_firstflrsf_med),
      `2ndFlrSF` = replace_na(`2ndFlrSF`, train_secondflrsf_med),
      BedroomAbvGr = replace_na(BedroomAbvGr, train_bedroomabvgr_med),
      GarageArea = replace_na(GarageArea, train_garagearea_med),
      PoolArea = replace_na(PoolArea, train_poolarea_med),
      GarageYrBlt = replace_na(GarageYrBlt, train_garageyrblt_med),
      YearBuilt = replace_na(YearBuilt, median(YearBuilt, na.rm =
TRUE)),
      YearRemodAdd = replace_na(YearRemodAdd, median(YearRemodAdd, na.rm =
TRUE)),
      OverallQual = replace_na(OverallQual, median(OverallQual, na.rm =
TRUE)),
      OverallCond = replace_na(OverallCond, median(OverallCond, na.rm =
TRUE)),
      Fireplaces = replace_na(Fireplaces, 0),
      MSSubClass = as.factor(replace_na(as.character(MSSubClass),
train_mssubclass_mode)),
      MSZoning = as.factor(replace_na(as.character(MSZoning),
train_mszoning_mode)),
      RoofMatl = as.factor(RoofMatl),
      Neighborhood = as.factor(Neighborhood),
      SaleCondition = as.factor(replace_na(as.character(SaleCondition),
train_salecond_mode)),
      Condition2 = as.factor(replace_na(as.character(Condition2),
train_condition2_mode)),
      LandSlope = as.factor(replace_na(as.character(LandSlope),
train_landslope_mode)),

```

```

    KitchenQual = as.factor(replace_na(as.character(KitchenQual),
train_kitchenqual_mode)),
    BsmtExposure = as.factor(replace_na(as.character(BsmtExposure),
train_bsmtexp_mode)),
    BsmtQual = as.factor(replace_na(as.character(BsmtQual),
train_bsmtqual_mode)),
    ExterQual = as.factor(replace_na(as.character(ExterQual),
train_extqual_mode)),
    Functional = as.factor(replace_na(as.character(Functional),
train_functional_mode)),
    GarageFinish = as.factor(replace_na(as.character(GarageFinish),
train_garagefinish_mode)),
    CentralAir = as.factor(CentralAir),
    TotalSF = TotalBsmtSF + `1stFlrSF` + `2ndFlrSF`,
    logTotalSF = log(TotalSF),
    logGrLivArea = log(GrLivArea),
    logLotArea = log(LotArea),
    TotalBath = FullBath + 0.5*HalfBath + BsmtFullBath + 0.5*BsmthalfBath,
    Age = YrSold - YearBuilt,
    RemodAge = YrSold - YearRemodAdd,
    GarageAge = YrSold - GarageYrBlt,
    TotalPorch = WoodDeckSF + OpenPorchSF + ScreenPorch,
    HasFireplace = as.factor(Fireplaces > 0),
    HasPool = as.factor(PoolArea > 0),
    Has2ndFloor = as.factor(`2ndFlrSF` > 0),
    OverallQual2 = OverallQual^2
  ) %>%
  left_join(neigh_lookup, by = "Neighborhood")
}

fix_unseen_levels <- function(test_df, train_df) {
  factor_cols <- names(test_df)[sapply(test_df, is.factor)]
  for (col in factor_cols) {
    if (col %in% names(train_df)) {
      train_levels <- levels(as.factor(train_df[[col]]))
      mode_val <- names(sort(table(train_df[[col]]), decreasing = TRUE))[1]
      test_df[[col]] <- as.character(test_df[[col]])
      test_df[[col]][!test_df[[col]] %in% train_levels] <- mode_val
      test_df[[col]] <- factor(test_df[[col]], levels = train_levels)
    }
  }
  test_df
}

# -----
# PREPROCESS
# -----
Train_full <- Train_raw %>% engineer() %>% mutate(logSalePrice = log(SalePrice))
Test <- Test_raw %>% engineer() %>% fix_unseen_levels(Train_full)

Train <- Train_full %>%
  filter(!(GrLivArea > 4000 & SalePrice < 200000))

cat(sprintf("Removed %d outliers. Training on %d rows.\n",
  nrow(Train_full) - nrow(Train), nrow(Train)))

# -----
# FIT MODELS
# -----
model_simple <- lm(logSalePrice ~ logTotalSF, data = Train)

model_reduced <- lm(logSalePrice ~
  logTotalSF + TotalBath + Has2ndFloor + TotalPorch +
  OverallQual + OverallCond + BsmtExposure + GarageFinish +
  CentralAir + FunctionalRed +
  Age + RemodAge +
  NeighborhoodGroup + MSZoning +

```

```

        GarageCars + HasFireplace + MasVnrArea + SaleCondition +
ScreenPorch +
        (NeighborhoodGroup * OverallQual) +
        (NeighborhoodGroup * logTotalSF),
        data = Train)

model_full <- lm(logSalePrice ~
        logTotalSF + logGrLivArea + logLotArea + TotalBath + Has2ndFloor
+ TotalPorch +
        OverallQual + OverallQual2 + OverallCond + ExterQual +
KitchenQual +
        BsmtQual + BsmtExposure + GarageFinish + CentralAir + Functional
+
        Age + RemodAge + GarageAge +
        Neighborhood + MSZoning +
        GarageCars + Fireplaces + HasFireplace + MasVnrArea +
SaleCondition + ScreenPorch +
        (Neighborhood * OverallQual) +
        (Neighborhood * logTotalSF),
        data = Train)

# -----
# PREDICT & EXPORT
# -----
save_submission <- function(test_data, model, filename) {
  test_data$SalePrice_pred <- exp(predict(model, newdata = test_data))
  na_count <- sum(is.na(test_data$SalePrice_pred))
  if (na_count > 0) warning(sprintf("%s: %d NA predictions!", filename, na_count))
  test_data %>%
    dplyr::select(Id, SalePrice = SalePrice_pred) %>%
    write_csv(path_(filename))
  cat(sprintf("Saved %s - price range $%.0f - $%.0f\n",
    filename,
    min(test_data$SalePrice_pred, na.rm = TRUE),
    max(test_data$SalePrice_pred, na.rm = TRUE)))
}

save_submission(Test, model_simple, "simple.csv")
save_submission(Test, model_reduced, "reduced.csv")
save_submission(Test, model_full, "full.csv")

# -----
# METRICS
# -----
press <- function(model) {
  h <- hatvalues(model); r <- residuals(model); valid <- h < 1
  sum((r[valid] / (1 - h[valid]))^2)
}

rmse_dollars <- function(model) {
  round(sqrt(mean((exp(model$model[[1]]) - exp(fitted(model)))^2)), 0)
}

metrics <- data.frame(
  Model = c("Simple", "Reduced", "Full"),
  Params = c(length(coef(model_simple)), length(coef(model_reduced)),
length(coef(model_full))),
  Adj_R2 = round(c(summary(model_simple)$adj.r.squared,
summary(model_reduced)$adj.r.squared, summary(model_full)$adj.r.squared), 4),
  CV_PRESS = c(press(model_simple), press(model_reduced), press(model_full)),
  AIC = round(c(AIC(model_simple), AIC(model_reduced), AIC(model_full)),
1),
  RMSE = c(rmse_dollars(model_simple), rmse_dollars(model_reduced),
rmse_dollars(model_full)),
  Kaggle_Score = c(0.22896, 0.13567, 0.13339)
)

metric_tbl <- metrics %>%

```

```

gt() %>%
tab_header(title = "Predictive Model Comparison") %>%
cols_label(
  Model = "Predictive Models", Params = "Parameters", Adj_R2 = "Adjusted R2",
  CV_PRESS = "CV PRESS", AIC = "AIC", RMSE = "RMSE ($)", Kaggle_Score = "Kaggle
Score"
) %>%
fmt_number(columns = Adj_R2, decimals = 4) %>%
fmt_number(columns = CV_PRESS, decimals = 2) %>%
fmt_number(columns = AIC, decimals = 1) %>%
fmt_number(columns = RMSE, decimals = 0) %>%
fmt_number(columns = Kaggle_Score, decimals = 5) %>%
tab_style(style = cell_fill(color = "#d4edda"),
  locations = cells_body(rows = Kaggle_Score == min(Kaggle_Score))) %>%
opt_row_stripping()

gtsave(metric_tbl, path_("plots/final_models.png"))

# -----
# DIAGNOSTIC PLOTS
# -----
make_diagnostics <- function(model, title) {
  df <- data.frame(
    fitted = fitted(model),
    residuals = residuals(model),
    stud_resid = rstudent(model),
    cooks = cooks.distance(model),
    obs = seq_len(nobs(model))
  ) %>%
  filter(is.finite(stud_resid))

  p1 <- ggplot(df, aes(fitted, stud_resid)) +
    geom_point(alpha = 0.3, size = 1) +
    geom_hline(yintercept = c(-2, 0, 2),
      linetype = c("dotted", "dashed", "dotted"),
      color = c("red", "red", "red")) +
    geom_smooth(method = "loess", se = FALSE, color = "steelblue") +
    labs(title = "Residuals vs Fitted", x = "Fitted Values", y = "Studentized
Residuals") +
    theme_minimal() + theme(plot.title = element_text(face = "bold"))

  p2 <- ggplot(df, aes(sample = stud_resid)) +
    stat_qq(alpha = 0.3, size = 1) + stat_qq_line(color = "red") +
    labs(title = "Normal Q-Q", x = "Theoretical Quantiles", y = "Studentized
Residuals") +
    theme_minimal() + theme(plot.title = element_text(face = "bold"))

  x_seq <- seq(min(df$stud_resid), max(df$stud_resid), length.out = 300)
  norm_df <- data.frame(
    x = x_seq,
    y = dnorm(x_seq, mean = mean(df$stud_resid), sd = sd(df$stud_resid))
  )

  p3 <- ggplot(df, aes(x = stud_resid)) +
    geom_histogram(aes(y = after_stat(density)), bins = 40,
      fill = "steelblue", color = "white", alpha = 0.7) +
    geom_line(data = norm_df, aes(x = x, y = y), color = "red", linewidth = 1) +
    labs(title = "Histogram of Residuals", x = "Studentized Residuals", y =
"Density") +
    theme_minimal() + theme(plot.title = element_text(face = "bold"))

  p4 <- ggplot(df, aes(obs, cooks)) +
    geom_col(alpha = 0.5, width = 1) +
    geom_hline(yintercept = 4 / nrow(df), linetype = "dashed", color = "red") +
    labs(title = "Cook's Distance", x = "Observation Index", y = "Cook's Distance")
+
  theme_minimal() + theme(plot.title = element_text(face = "bold"))

```

```
(p1 + p2) / (p3 + p4) +  
  plot_annotation(title = title, theme = theme(plot.title = element_text(face =  
"bold", size = 18, hjust = 0.5)))  
}  
  
make_diagnostics(model_full, "Full Model Diagnostic Plots")  
ggsave(paste0(project, "full_model_diagnostics.png"))  
make_diagnostics(model_reduced, "Reduced Model Diagnostic Plots")  
ggsave(paste0(project, "red_model_diagnostics.png"))
```

```

###AMES MAP

library(AmesHousing)
library(ggplot2)
library(dplyr)
library(sf)
library(concaveman)
library(ggrepel)
library(patchwork)

ames_full <- make_ames()

# --- Build clusters directly from AmesHousing data ---
neigh_base2 <- ames_full %>%
  mutate(log_price = log(Sale_Price)) %>%
  group_by(Neighborhood) %>%
  summarise(mean_log_price = mean(log_price, na.rm = TRUE), .groups = "drop")

set.seed(123)
km2 <- kmeans(neigh_base2[, "mean_log_price", drop = FALSE], centers = 10, nstart =
25)
neigh_base2$cluster <- km2$cluster

cluster_order2 <- neigh_base2 %>%
  group_by(cluster) %>%
  summarise(cluster_mean = mean(mean_log_price), .groups = "drop") %>%
  arrange(cluster_mean)

label_map2 <- setNames(paste0("Group_", 1:10), cluster_order2$cluster)
neigh_base2$NeighborhoodGroup <-
  factor(label_map2[as.character(neigh_base2$cluster)],
         levels = paste0("Group_", 1:10))

# --- Join back to full data ---
ames_plot <- ames_full %>%
  left_join(neigh_base2 %>% select(Neighborhood, NeighborhoodGroup,
mean_log_price),
           by = "Neighborhood")

# --- Build concave hull polygons ---
neigh_list <- split(ames_plot, ames_plot$Neighborhood)
hull_list <- lapply(names(neigh_list), function(nm) {
  df <- neigh_list[[nm]]
  if (nrow(df) < 3) return(NULL)
  pts <- st_as_sf(df, coords = c("Longitude", "Latitude"), crs = 4326)
  hull <- concaveman(pts, concavity = 2)
  hull$Neighborhood <- nm
  hull
})
neigh_polygons <- do.call(rbind, hull_list[!sapply(hull_list, is.null)]) %>%
  left_join(neigh_base2 %>% select(Neighborhood, NeighborhoodGroup,
mean_log_price),
           by = "Neighborhood")

# --- Clean up labels ---
centroids <- centroids %>%
  mutate(label = gsub("_", " ", Neighborhood))

# --- Name mapping: full names -> abbreviations ---
neigh_abbrev <- tribble(
  ~Neighborhood, ~abbr,
  "Bloomington_Heights", "Blmngtn",
  "Bluestem", "Blueste",
  "Briardale", "BrDale",
  "Brookside", "BrkSide",
  "Clear_Creek", "ClearCr",
  "College_Creek", "CollgCr",
  "Crawford", "Crawfor",

```

```

"Edwards",
"Gilbert",
"Greens",
"Green_Hills",
"Iowa_DOT_and_Rail_Road",
"Landmark",
"Meadow_Village",
"Mitchell",
"North_Ames",
"Northridge",
"Northpark_Villa",
"Northridge_Heights",
"Northwest_Ames",
"Old_Town",
"South_and_West_of_Iowa_State_University",
"Sawyer",
"Sawyer_West",
"Somerset",
"Stone_Brook",
"Timberland",
"Veenker",
)

"Edwards",
"Gilbert",
"Greens",
"GrnHill",
"IDOTRR",
"Landmrk",
"MeadowV",
"Mitchel",
"NAmes",
"NoRidge",
"NPKvill",
"NridgHt",
"NWAmes",
"OldTown",
"SWISU",
"Sawyer",
"SawyerW",
"Somerst",
"StoneBr",
"Timber",
"Veenker"

)

centroids <- centroids %>%
  left_join(neigh_abbrev, by = "Neighborhood")

# --- Plot 1: by mean price ---
p1 <- ggplot() +
  geom_sf(data = neigh_polygons, aes(fill = mean_log_price),
    color = "white", linewidth = 0.5, alpha = 0.7) +
  geom_text_repel(data = centroids, aes(x = lon, y = lat, label = abbr),
    size = 3.5, max.overlaps = 30, segment.size = 0.3,
    fontface = "bold") +
  scale_fill_viridis_c(option = "C", name = "Mean\nlog(Price)") +
  labs(title = "Ames Neighborhoods by Mean log(SalePrice)") +
  theme_excel_new() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.title = element_text(size = 12),
    legend.position = "right",
    legend.text = element_text(size = 12),
    axis.title = element_blank())

# --- Plot 2: Bar chart of house counts by neighborhood ---
neigh_counts <- ames_plot %>%
  count(Neighborhood) %>%
  left_join(neigh_abbrev, by = "Neighborhood") %>%
  mutate(abbr = ifelse(is.na(abbr), Neighborhood, abbr))

p2 <- ggplot(neigh_counts, aes(x = reorder(abbr, n), y = n, fill = abbr)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = n), hjust = -0.15, size = 3) +
  coord_flip() +
  labs(title = "Number of Houses Sold by Neighborhood",
    x = NULL, y = "Count") +
  theme_excel_new() +
  theme(plot.title = element_text(face = "bold", size = 14),
    axis.text.y = element_text(size = 9))

p1 + p2

ggsave('/Users/jwkle/Documents/Stats/HW/project/plots/Ames-neighborhoods.png')
```